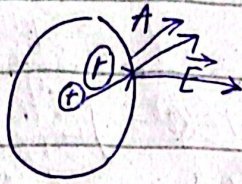


Relation of Coulomb's law and Gauss's law



we have to find E.F at this point due to positive charge inside the sphere.

Apply Gauss's law

$$\begin{aligned}\phi &= \int E \cdot dA = \int E dA \cos \theta \quad \because \theta = 0^\circ \\ &= E \int dA \\ &= E 4\pi r^2 \quad \because A = 4\pi r^2\end{aligned}$$

Gauss's 2nd eq.

$$\phi = Q/\epsilon_0$$

Compare eq. (i) and (ii)

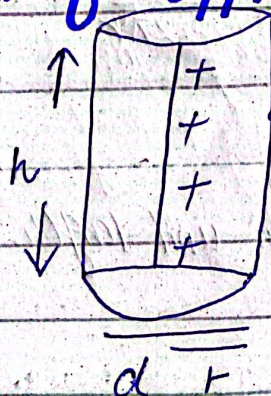
$$4\pi r^2 E = Q/\epsilon_0$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

Same information as Coulomb's law.

Derivation of application:

The cylinder is gaussian surface.



This point we want to find E.F.

Here we consider Symmetrical charge distribution on infinite wire

$$\phi_{\text{top side}} = \int E \cdot dA = \int 0 dA = 0$$

As on top and bottom side, there is zero E.F area.

$\phi_{\text{bottom side}}$ = same case as top

$$\begin{aligned}\phi_{\text{curre}} &= \int E dA \\ &= \int E dA \cos \theta \\ &= E \int dA \\ &= 2\pi r l E\end{aligned}$$

Gauss's 2nd equation

$$\phi = \frac{q}{\epsilon_0}$$

Compare eq. (i) and (ii)

$$E 2\pi r l = q / \epsilon_0$$

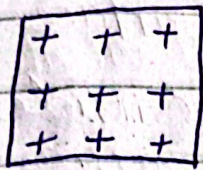
$$E = \frac{q}{\epsilon_0 2\pi r l}$$

$$E = \frac{\lambda}{\epsilon_0 2\pi r}$$

Linear charge density $\lambda = \frac{q}{l}$

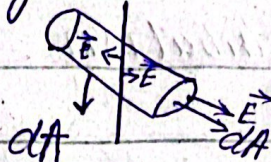
which tells how much charge is present in unit length.

Planar Symmetry:



→ consider symmetrical charge distribution on infinite sheet

Again we consider Gaussian surface 'cylinder' and pass through sheet



$E \cdot F$ at curve side is zero. We get fun at front and rear side only.

Apply Gauss's law first eq.

$$\begin{aligned}\phi_f &= \int E dA \\ &= \int E dA \cos \alpha\end{aligned}$$

$$\alpha = 0^\circ$$

$$\begin{aligned}&= E \int dA \\ &= EA\end{aligned}$$

$$\phi_r = EA$$

$$\phi_{net} = 2EA$$

$$\frac{q}{\epsilon_0} = 2EA$$

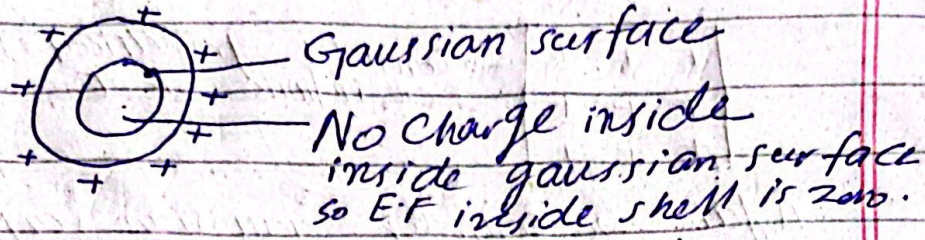
$$\sigma = \frac{q}{A} \text{ which tell}$$

how much charge is present in unit area.

$$E = \frac{q}{2A\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

Spherical symmetry:



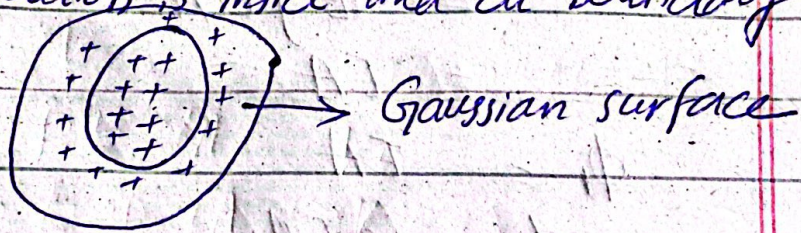
Consider a shell, there is symmetrical charge distribution outside the boundary.

According to shell theory, it behave like a single charge. $E \cdot F$ will be same as of point charge.

$$E \cdot F = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

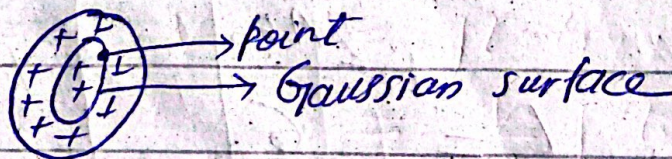
$$E = 0$$

Now, we consider any shell in which charge distribution is inside and at boundary as well.



$$E \cdot F = \frac{1}{4\pi\epsilon_0} \frac{q'}{r^2} \rightarrow \text{charge inside gaussian surface}$$

If point is consider inside shell



We find charge inside by using charge to volume ratio.

$$\frac{qV'}{\frac{4\pi}{3} r^3} = \frac{qV}{\frac{4\pi}{3} R^3}$$

$$qV' = \frac{qV r^3}{R^3}$$

Put this in E.F formula

$$E.F = \frac{1}{4\pi\epsilon_0} \frac{qV'}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{qV r^3}{R^3 r^2}$$

$$= \frac{qV R}{4\pi\epsilon_0 R^3}$$

Chap 25

$$Q \propto V$$

$$Q = \underset{\substack{\downarrow \\ \text{always constant}}}{C} V$$

Checkpoint 1 $\rightarrow$ (a) same (b) same

calculate capacitance

1 charge q

2 use gauss's law to calculate $\vec{E}$

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

3 Potential gradient $E = -\frac{\Delta V}{\Delta r}$

$$V = -\int \vec{E} \cdot d\vec{s}$$

4 $C = \frac{Q}{V}$

parallel plate

q

$$\epsilon_0 \oint \vec{E} \cdot d\vec{A} = q$$

$$\epsilon_0 E \oint dA = q$$

$$\epsilon_0 EA = q$$

$$E = \frac{q}{\epsilon_0 A}$$

$$V = -\int \vec{E} \cdot d\vec{s} = -\int E ds \cos 180^\circ$$

$$= \int E ds$$

$$= \frac{q}{\epsilon_0 A} \int ds$$

$$V = \frac{q}{\epsilon_0 A} d$$

$$C = \frac{\epsilon_0 A q}{q d} = \frac{\epsilon_0 A}{d}$$

depends upon Area and distance

Cylinder:

qV

$$\epsilon_0 \oint E \cdot dA = qV$$

$$\epsilon_0 E \oint dA = qV \quad \because A = 4\pi r^2$$

$$E = \frac{qV}{\epsilon_0 4\pi r^2}$$

$$V = - \int \vec{E} \cdot d\vec{s}$$

$$= \int E ds$$

$$= \frac{qV}{\epsilon_0 4\pi r^2} \int_b^a ds \quad \because ds = -dr$$

$$= - \frac{qV}{\epsilon_0 4\pi} \int_b^a \frac{dr}{r^2}$$

$$= - \frac{qV}{\epsilon_0 4\pi} \left[\frac{r^{-2+1}}{-1} \right]_b^a$$

$$= - \frac{qV}{\epsilon_0 4\pi} \left(\frac{a^{-1}}{-1} - \frac{b^{-1}}{-1} \right)$$

$$= - \frac{qV}{\epsilon_0 4\pi} \left(\frac{-1}{a} + \frac{1}{b} \right)$$

$$= - \frac{qV}{\epsilon_0 4\pi} \left(\frac{-b+a}{ab} \right)$$

$$= - \frac{qV}{\epsilon_0 4\pi} \left(\frac{a-b}{ab} \right) = \frac{qV}{4\pi \epsilon_0} \left(\frac{b-a}{ab} \right)$$

$C = \frac{Q}{V}$
Cylindrical capacitor:

$$\phi = \frac{qV}{\epsilon_0}$$

$$\phi = \int E \cdot A$$

$$\epsilon_0 \int E \cdot A = qV$$

$$\epsilon_0 E (2\pi r l) = qV$$

$$E = \frac{qV}{\epsilon_0 2\pi r l}$$

$$V = - \int \vec{E} \cdot d\vec{s}$$

$$= - \int E ds \cos 180^\circ \quad ds = -dr$$

$$= \int E ds$$

$$= \frac{qV}{\epsilon_0 2\pi r l} ds$$

$$= \frac{-qV}{\epsilon_0 2\pi l} \int \frac{dr}{r} = \frac{-qV}{\epsilon_0 2\pi l} \int_b^a \frac{dr}{r}$$

$$= \frac{-qV}{2\epsilon_0 \pi l} \ln r \Big|_b^a$$

$$= \frac{-qV}{2\pi \epsilon_0 l} (\ln a - \ln b)$$

$$= \frac{qV}{2\pi \epsilon_0 l} (\ln b - \ln a)$$

$$= \frac{qV}{2\pi \epsilon_0 l} \ln\left(\frac{b}{a}\right)$$

$$\phi = CV$$

$$C = \frac{Q}{V} = \frac{Q \cdot 2\pi \epsilon_0 l}{qV \ln(b/a)}$$

$$= \frac{2\pi \epsilon_0 l}{\ln(b/a)}$$

Chap 29

$$dB \propto i$$

$$dB \propto ds$$

$$dB \propto \frac{1}{r^2}$$

$$dB \propto \frac{id\vec{s}}{r^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{id\vec{s}}{r^2} \sin \alpha$$

Biot and Savart

$$dB = \frac{\mu_0}{4\pi} \frac{id\vec{s}}{r^2} \hat{r}$$

$$= \frac{\mu_0}{4\pi} \frac{id\vec{s} \vec{r}}{r^3}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$$

Complete wire $\rightarrow \frac{\mu_0 i}{2\pi R}$

$$F_a = i_b l B_a \sin \alpha = \frac{\mu_0 L i_a i_b}{2\pi d}$$

$$B = \frac{\mu_0 i}{2\pi r}$$

$$\leftarrow \otimes \rightarrow \quad \leftarrow \otimes \rightarrow \quad \otimes$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i$$

$$= \int B ds \cos 0$$

$$= B \int ds$$

$$\mu_0 i = B 2\pi r$$

$$B = \frac{\mu_0 i}{2\pi r}$$

derivation

Solenoid

$$\begin{aligned} &= \int \vec{B} \cdot d\vec{s} \\ &= \int_a^b \vec{B} \cdot d\vec{s} + \int_b^c \vec{B} \cdot d\vec{s} + \int_c^d \vec{B} \cdot d\vec{s} \\ &\quad + \int_d^a \vec{B} \cdot d\vec{s} \end{aligned}$$

$$\begin{aligned} &= \int_a^b B ds \cos 0^\circ + \int_b^c B ds \cos 90^\circ \\ &\quad + \int_c^d B ds \cos 0^\circ + \int_d^a B ds \cos 90^\circ \end{aligned}$$

$\downarrow$
negligible

$$= \int_a^b B ds$$

$$= B l$$

$$i_{enc} = n l i$$

$$\mu_0 i_{enc} = B l$$

$$\mu_0 n l i = B l$$

$$B = \mu_0 n i$$